Supplementary File

This file consists of some results of paper "Graph-Based Minimally Supervised Risk Stratification for Prognosis Prediction in Heart Failure with preserved Ejection Fraction"

1. DATA ACQUISITION

A. Circulating Biomarkers and Proteomics:

Blood Samples including renal function and hematocrit were taken at the time of recruitment. Circulating biomarkers were quantified using a bead-based multiplex assay on a Luminex platform (Bristol Myers Squibb, NJ, USA) as described in [1]. This includes 49 biomarkers associated with fibrosis, myocardial injury, atrial stretch, myocardial hypertrophy, inflammation and oxidative stress, renal markers and endothelial dysfunction.

Proteomics analysis in the DIAMOND-HFpEF study was conducted on plasma samples collected from patients. The Extracellular Vesicles (ECV) and Lipid Removal Agent (LRA) protocol were employed for isolation of extracellular vesicles and low-abundant proteins on affinity matrices before analysing on mass spectrometry in the Leicester van Geest Multi-Omics Facility. Briefly, 200 µL of plasma was used to isolate ECV proteins using precipitation of ECV with ammonium acetate pH 4.75 as described in [2], and the supernatant was carefully collected for analysis of low-abundance proteins using the LRA protocol [3]. Proteins were subjected to reduction with DL-dithiothreitol (DTT) to break disulphide bonds, followed by alkylation with iodoacetamide (IAA) to prevent reformation of these bonds. Proteolytic digestion was then performed using trypsin at a 1:25 enzyme-to-protein ratio. The peptides were purified using EMPORE C18 discs with 60% and then 80% acetonitrile (ACN). Solvent was removed using a speed vacuum concentrator and samples were lyophilised overnight. Peptides were resuspended in 0.1% FA, and peptide concentrations were determined using a peptide assay prior to mass spectrometry. Mass spectrometric analysis was performed on a Waters Synapt G2S High Definition Mass Spectrometry (HDMS) system including a nano-electrospray ionisation (ESI) source, stepwave ion guide, quadrupole, triwave ion mobility, and time-of-flight (TOF) analyser that was coupled to a nanoAcquity Ultra-Performance Liquid Chromatography (UPLC) system for peptide separation. Each sample was analysed using a 75-minute LC gradient in HDMSE acquisition mode in LC-ESI-MS/MS.

2. MATERIAL AND METHOD

A. Patient Graph Construction:

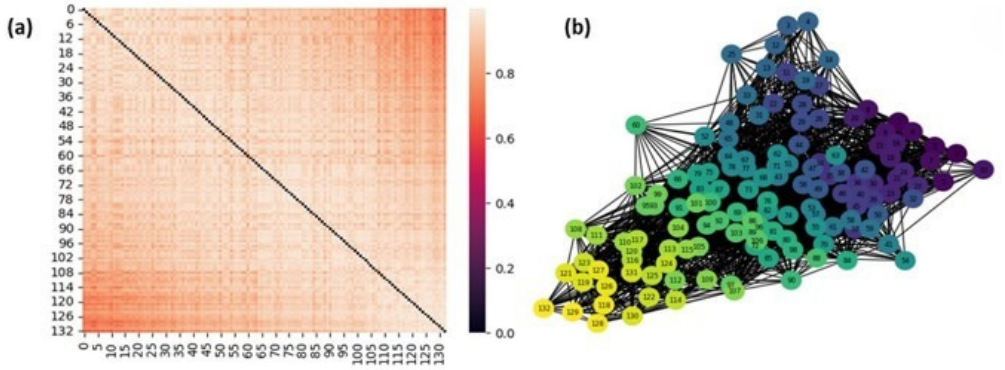


Fig. S1. Patient Graph Construction: (a) Cosine similarity of 133 patients based on selected clinical features. (b) Graph constructed by cosine similarity of selected clinical features.

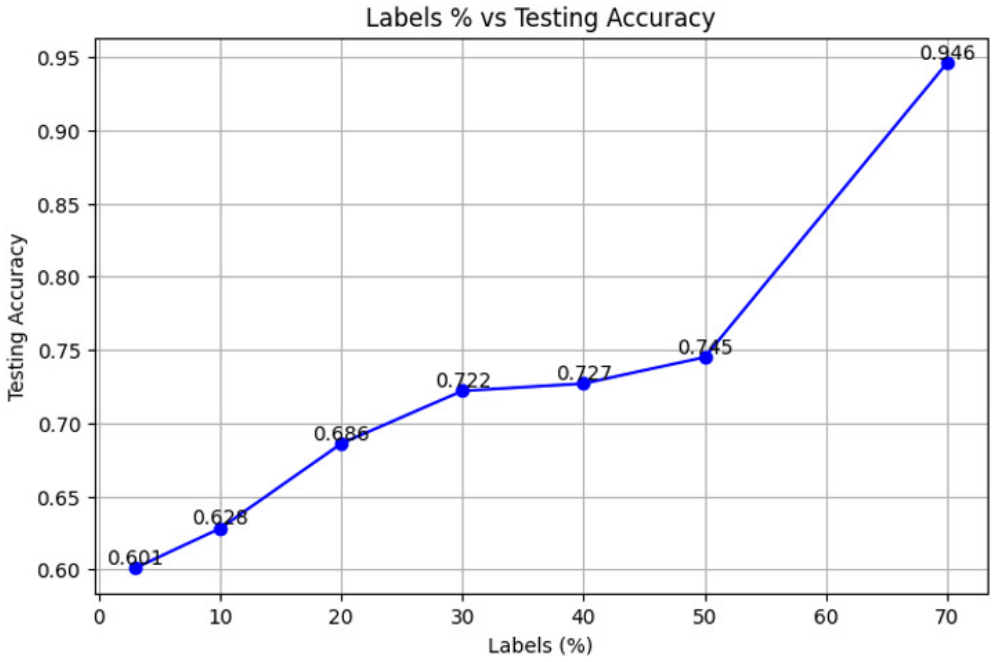


Fig. S2. label-rate sensitivity analysis

3. EXPERIMENTAL SETUP

4. RESULTS AND DISCUSSION

A. Survival and Statistical Analysis:

- Box plots are commonly used to visualise the distribution of data and to compare different electrocardiographic features in different groups. Several clinical variables show overlapping interquartile ranges across the GCNN-derived clusters. This is expected because the clusters are not formed by separating patients along any single variable. Instead, the GCNN identifies groups based on multivariate patterns, capturing joint feature interactions, nonlinear relationships, and patient-to-patient similarity encoded through the cosine-similarity graph. As a result, clusters may appear similar when examined using individual box plots, yet differ substantially in the high-dimensional feature space learned by the model. Such overlap in single-variable distributions is common in clinical datasets due to biological heterogeneity and may coexist with meaningful subgroup structure that becomes apparent only when multiple features are considered jointly.

- Tables S1 and S2 display the results of the ANOVA test and pairwise comparison.

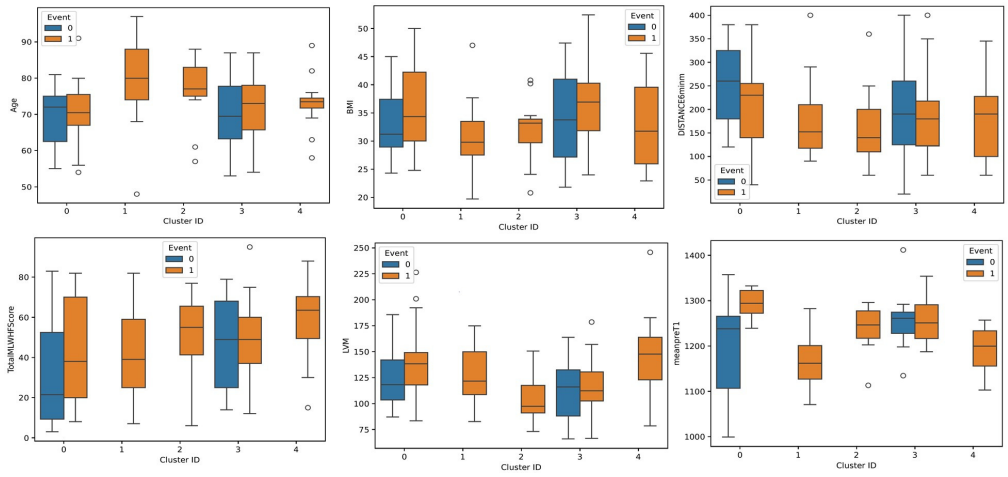


Fig. S3. Box plots for multi-modal HFpEF dataset.

Table S1. ANCOVA results

	Mean (SD)						Adjusted Mean					
Biomarker	C0	C1	C2	C3	C4	Unadj p	C0	C1	C2	C3	C4	Adj p
log10_Angiopoietin2	3.29 (0.23)	3.41 (0.17)	3.30 (0.20)	3.31 (0.21)	3.30 (0.32)	0.368	3.30	3.42	3.31	3.30	3.30	0.338
log10_EndoglinCD105	2.62 (0.49)	2.72 (0.77)	2.87 (0.15)	2.60 (0.62)	2.50 (0.80)	0.513	2.68	2.70	2.92	2.56	2.49	0.308
log10_FABP3	2.46 (0.05)	2.45 (0.00)	2.45 (0.00)	2.42 (0.36)	2.45 (0.00)	0.924	2.47	2.46	2.46	2.41	2.45	0.825
log10_ICAM_1	5.14 (0.24)	5.04 (1.19)	5.30 (0.25)	5.10 (0.79)	5.14 (0.45)	0.893	5.12	5.07	5.30	5.10	5.15	0.919
log10_OPG	1.98 (0.11)	2.05 (0.08)	2.06 (0.17)	1.97 (0.17)	2.05 (0.16)	0.047	1.99	2.05	2.07	1.96	2.05	0.055
log10_Osteopontin	4.34 (0.23)	4.43 (0.24)	4.45 (0.19)	4.23 (0.65)	4.42 (0.27)	0.268	4.36	4.43	4.47	4.21	4.42	0.196
log10_KIM1	-0.26 (0.00)	0.02 (0.65)	-0.26 (0.00)	-0.21 (0.28)	-0.08 (0.72)	0.085	-0.27	0.00	-0.29	-0.18	-0.08	0.114
log10_VEGFR1	0.91 (0.34)	1.12 (0.22)	1.08 (0.12)	0.90 (0.31)	0.88 (0.41)	0.023	0.91	1.12	1.09	0.90	0.88	0.028
log10_Endothelin1	0.27 (0.09)	0.26 (0.00)	0.26 (0.00)	0.26 (0.00)	0.26 (0.00)	0.559	0.27	0.26	0.26	0.26	0.26	0.657
log10_FABP4	3.97 (0.57)	3.91 (0.69)	4.09 (0.30)	3.97 (0.47)	3.94 (0.44)	0.908	3.96	4.00	4.15	3.92	3.97	0.724
log10_IL1beta	-0.41 (0.26)	-0.40 (0.29)	-0.45 (0.02)	-0.38 (0.34)	-0.32 (0.42)	0.779	-0.41	-0.40	-0.45	-0.39	-0.32	0.836
log10_IL10	-1.74 (0.27)	-1.60 (0.41)	-1.54 (0.53)	-1.58 (0.63)	-1.61 (0.53)	0.649	-1.75	-1.63	-1.57	-1.56	-1.62	0.576
log10_IL6	0.61 (0.14)	0.66 (0.19)	0.62 (0.07)	0.62 (0.18)	0.63 (0.15)	0.808	0.60	0.67	0.62	0.62	0.63	0.628
log10_IL8	0.50 (0.46)	0.42 (0.50)	0.58 (0.29)	0.41 (0.50)	0.52 (0.39)	0.683	0.52	0.41	0.59	0.40	0.52	0.605
log10_PentraxinC	1.77 (0.56)	2.10 (0.61)	1.99 (0.70)	1.65 (0.51)	2.15 (0.67)	0.005	1.77	2.10	1.98	1.65	2.15	0.010
log10_Renin	2.69 (0.41)	2.64 (0.40)	2.69 (0.42)	2.60 (0.50)	2.58 (0.40)	0.850	2.66	2.65	2.68	2.61	2.59	0.960
log10_TNF_RI	3.72 (0.17)	3.79 (0.21)	3.77 (0.17)	3.74 (0.19)	3.80 (0.23)	0.532	3.72	3.80	3.77	3.74	3.81	0.356
log10_TNFalpha	-0.37 (0.43)	-0.11 (0.63)	-0.50 (0.25)	-0.43 (0.39)	-0.33 (0.49)	0.078	-0.36	-0.09	-0.48	-0.44	-0.32	0.050
log10_proBNP	0.24 (0.21)	0.26 (0.14)	0.22 (0.22)	0.22 (0.19)	0.28 (0.14)	0.719	0.25	0.24	0.21	0.22	0.28	0.819

	Mean (SD)						Adjusted Mean					
Biomarker	C0	C1	C2	C3	C4	Unadj p	C0	C1	C2	C3	C4	Adj p
log10_VEGF	1.78 (0.14)	1.79 (0.12)	1.74 (0.18)	1.72 (0.16)	1.75 (0.13)	0.257	1.78	1.79	1.74	1.71	1.75	0.266
log10_Chitinase3Like1	4.52 (0.38)	4.59 (0.37)	4.68 (0.35)	4.49 (0.37)	4.48 (0.83)	0.682	4.54	4.61	4.70	4.47	4.48	0.487
log10_Endostatin	5.18 (0.14)	5.26 (0.17)	5.27 (0.17)	5.19 (0.18)	5.12 (0.48)	0.262	5.18	5.27	5.28	5.18	5.12	0.201
log10_Fas	4.01 (0.10)	4.05 (0.12)	4.05 (0.17)	3.98 (0.18)	3.96 (0.36)	0.474	4.01	4.05	4.05	3.98	3.96	0.488
log10_Pselectin	4.46 (0.19)	4.44 (0.16)	4.49 (0.22)	4.39 (0.16)	4.28 (0.50)	0.084	4.46	4.45	4.49	4.38	4.28	0.076
log10_TenascinC	4.13 (0.14)	4.22 (0.14)	4.18 (0.14)	4.12 (0.14)	4.10 (0.25)	0.107	4.14	4.22	4.19	4.12	4.10	0.116
log10_Adiponectin	7.79 (0.22)	7.84 (0.23)	7.81 (0.28)	7.71 (0.30)	7.52 (1.12)	0.240	7.81	7.81	7.81	7.71	7.51	0.219
log10_SerpinE1_PAI1	5.36 (0.31)	5.26 (0.27)	5.42 (0.36)	5.24 (0.25)	5.12 (0.43)	0.033	5.35	5.30	5.44	5.22	5.13	0.027
log10_TNFRII	2.16 (1.27)	2.45 (1.36)	2.61 (1.21)	2.19 (1.33)	2.53 (1.47)	0.706	2.14	2.51	2.64	2.16	2.55	0.591
log10_TIMP4	3.27 (0.16)	3.32 (0.16)	3.33 (0.13)	3.13 (0.49)	3.15 (0.54)	0.165	3.29	3.33	3.37	3.09	3.16	0.040
log10_MMP12	1.19 (0.21)	1.25 (0.14)	1.08 (0.22)	1.15 (0.26)	1.14 (0.28)	0.213	1.21	1.25	1.08	1.15	1.13	0.217
log10_MMP7	2.79 (0.38)	2.91 (0.34)	2.66 (0.62)	2.77 (0.32)	3.03 (0.36)	0.050	2.78	2.91	2.65	2.77	3.03	0.057
log10_MMP8	2.48 (0.24)	2.54 (0.24)	2.51 (0.24)	2.43 (0.21)	2.43 (0.33)	0.345	2.46	2.57	2.52	2.42	2.44	0.220
log10_MMP2	4.85 (0.09)	4.93 (0.10)	4.85 (0.12)	4.83 (0.10)	4.88 (0.15)	0.017	4.86	4.92	4.85	4.83	4.87	0.045
log10_MMP3	3.90 (0.24)	3.92 (0.23)	3.87 (0.14)	3.75 (0.23)	3.82 (0.40)	0.037	3.89	3.91	3.86	3.76	3.82	0.118
log10_MMP9	4.54 (0.29)	4.53 (0.31)	4.62 (0.35)	4.48 (0.28)	4.34 (0.65)	0.236	4.53	4.56	4.63	4.47	4.35	0.235
log10_Timp1	6.00 (0.15)	6.05 (0.19)	6.06 (0.24)	5.97 (0.18)	5.99 (0.38)	0.519	6.00	6.07	6.08	5.96	5.99	0.203
log10_NTProANP	3.76 (0.21)	3.93 (0.12)	3.77 (0.22)	3.72 (0.21)	3.78 (0.27)	0.007	3.77	3.90	3.76	3.73	3.77	0.041
log10_ST2	3.80 (0.17)	3.82 (0.13)	3.82 (0.15)	3.79 (0.20)	3.82 (0.40)	0.977	3.80	3.81	3.81	3.80	3.82	0.994
log10_CystatinC	5.89 (0.09)	5.97 (0.11)	5.93 (0.08)	5.88 (0.10)	5.86 (0.34)	0.124	5.89	5.98	5.94	5.88	5.86	0.079
log10_GDF15	3.35 (0.27)	3.49 (0.21)	3.49 (0.29)	3.29 (0.28)	3.51 (0.44)	0.012	3.34	3.49	3.47	3.30	3.51	0.025

	Mean (SD)						Adjusted Mean					
Biomarker	C0	C1	C2	C3	C4	Unadj p	C0	C1	C2	C3	C4	Adj p
log10_Galectin3	3.87 (0.12)	3.89 (0.14)	3.91 (0.13)	3.83 (0.12)	3.88 (0.12)	0.141	3.86	3.90	3.91	3.83	3.88	0.203
log10_NGAL	4.64 (0.13)	4.73 (0.18)	4.68 (0.15)	4.61 (0.18)	4.63 (0.31)	0.119	4.64	4.74	4.68	4.61	4.63	0.149
log10_MPO	5.33 (0.19)	5.34 (0.15)	5.32 (0.19)	5.29 (0.14)	5.25 (0.35)	0.563	5.33	5.35	5.33	5.28	5.25	0.413
log10_FGF_23	1.83 (0.30)	2.05 (0.28)	2.00 (0.29)	1.75 (0.40)	1.74 (0.52)	0.011	1.82	2.09	2.03	1.72	1.76	0.001
log10_FGF21	0.80 (1.02)	0.60 (0.84)	0.49 (0.60)	0.33 (0.42)	0.61 (0.90)	0.095	0.76	0.62	0.46	0.36	0.62	0.216
log10_Syndecan1	2.73 (0.15)	2.77 (0.16)	2.71 (0.17)	2.65 (0.42)	2.79 (0.31)	0.346	2.74	2.78	2.72	2.64	2.79	0.274
log10_Syndecan4	2.55 (0.29)	2.43 (0.21)	2.61 (0.31)	2.35 (0.42)	2.49 (0.32)	0.034	2.57	2.45	2.64	2.32	2.49	0.008

Table S2. Pairwise comparisons of biomarkers across clusters

Biomarker	Comparison	Difference	SE	t	p_raw	p_bonf	Sig
log10_VEGFR1	1 vs 0	0.21	0.09	2.37	0.019	0.191	
log10_VEGFR1	2 vs 0	0.17	0.10	1.72	0.087	0.873	
log10_VEGFR1	3 vs 0	-0.02	0.07	-0.21	0.832	1	
log10_VEGFR1	4 vs 0	-0.03	0.09	-0.34	0.731	1	
log10_VEGFR1	2 vs 1	-0.03	0.11	-0.32	0.753	1	
log10_VEGFR1	3 vs 1	-0.22	0.08	-2.71	0.008	0.076	
log10_VEGFR1	4 vs 1	-0.24	0.10	-2.36	0.020	0.199	
log10_VEGFR1	3 vs 2	-0.19	0.10	-1.92	0.057	0.572	
log10_VEGFR1	4 vs 2	-0.21	0.12	-1.79	0.075	0.754	
log10_VEGFR1	4 vs 3	-0.02	0.09	-0.19	0.847	1	
log10_PentraxinC	1 vs 0	0.33	0.17	1.98	0.050	0.499	
log10_PentraxinC	2 vs 0	0.21	0.19	1.10	0.272	1	
log10_PentraxinC	3 vs 0	-0.11	0.14	-0.82	0.416	1	
log10_PentraxinC	4 vs 0	0.38	0.18	2.13	0.035	0.353	
log10_PentraxinC	2 vs 1	-0.12	0.21	-0.57	0.568	1	
log10_PentraxinC	3 vs 1	-0.44	0.16	-2.82	0.006	0.056	
log10_PentraxinC	4 vs 1	0.05	0.19	0.27	0.791	1	
log10_PentraxinC	3 vs 2	-0.32	0.19	-1.72	0.087	0.872	
log10_PentraxinC	4 vs 2	0.17	0.22	0.78	0.438	1	
log10_PentraxinC	4 vs 3	0.49	0.17	2.91	0.004	0.043	*
log10_SerpinE1_PAI1	1 vs 0	-0.04	0.08	-0.51	0.608	1	
log10_SerpinE1_PAI1	2 vs 0	0.09	0.09	0.99	0.325	1	
log10_SerpinE1_PAI1	3 vs 0	-0.12	0.07	-1.81	0.072	0.722	
log10_SerpinE1_PAI1	4 vs 0	-0.21	0.09	-2.43	0.017	0.166	
log10_SerpinE1_PAI1	2 vs 1	0.14	0.10	1.33	0.186	1	
log10_SerpinE1_PAI1	3 vs 1	-0.08	0.08	-1.03	0.304	1	
log10_SerpinE1_PAI1	4 vs 1	-0.17	0.10	-1.80	0.074	0.739	
log10_SerpinE1_PAI1	3 vs 2	-0.22	0.09	-2.33	0.021	0.212	
log10_SerpinE1_PAI1	4 vs 2	-0.31	0.11	-2.86	0.005	0.050	
log10_SerpinE1_PAI1	4 vs 3	-0.09	0.08	-1.11	0.269	1	
log10_TIMP4	1 vs 0	0.03	0.11	0.33	0.742	1	
log10_TIMP4	2 vs 0	0.08	0.12	0.63	0.531	1	
log10_TIMP4	3 vs 0	-0.20	0.09	-2.30	0.023	0.229	
log10_TIMP4	4 vs 0	-0.14	0.11	-1.21	0.229	1	
log10_TIMP4	2 vs 1	0.04	0.13	0.32	0.752	1	
log10_TIMP4	3 vs 1	-0.23	0.10	-2.36	0.020	0.199	
log10_TIMP4	4 vs 1	-0.17	0.12	-1.40	0.164	1	

Biomarker	Comparison	Difference	SE	t	p_raw	p_bonf	Sig
log10_TIMP4	3 vs 2	-0.27	0.12	-2.32	0.022	0.218	
log10_TIMP4	4 vs 2	-0.21	0.14	-1.54	0.126	1	
log10_TIMP4	4 vs 3	0.06	0.11	0.58	0.565	1	
log10_MMP2	1 vs 0	0.06	0.03	2.05	0.042	0.421	
log10_MMP2	2 vs 0	-0.01	0.04	-0.25	0.802	1	
log10_MMP2	3 vs 0	-0.03	0.03	-1.04	0.301	1	
log10_MMP2	4 vs 0	0.02	0.03	0.52	0.601	1	
log10_MMP2	2 vs 1	-0.07	0.04	-1.89	0.062	0.617	
log10_MMP2	3 vs 1	-0.09	0.03	-3.09	0.002	0.025	*
log10_MMP2	4 vs 1	-0.05	0.04	-1.28	0.204	1	
log10_MMP2	3 vs 2	-0.02	0.03	-0.50	0.618	1	
log10_MMP2	4 vs 2	0.03	0.04	0.65	0.516	1	
log10_MMP2	4 vs 3	0.04	0.03	1.39	0.166	1	
log10_NTProANP	1 vs 0	0.14	0.06	2.32	0.022	0.218	
log10_NTProANP	2 vs 0	-0.01	0.07	-0.18	0.858	1	
log10_NTProANP	3 vs 0	-0.04	0.05	-0.78	0.434	1	
log10_NTProANP	4 vs 0	0.01	0.06	0.08	0.936	1	
log10_NTProANP	2 vs 1	-0.15	0.07	-2.03	0.044	0.440	
log10_NTProANP	3 vs 1	-0.17	0.05	-3.15	0.002	0.020	*
log10_NTProANP	4 vs 1	-0.13	0.07	-1.92	0.057	0.572	
log10_NTProANP	3 vs 2	-0.03	0.07	-0.39	0.698	1	
log10_NTProANP	4 vs 2	0.02	0.08	0.22	0.824	1	
log10_NTProANP	4 vs 3	0.04	0.06	0.72	0.474	1	
log10_GDF15	1 vs 0	0.16	0.08	1.88	0.062	0.622	
log10_GDF15	2 vs 0	0.14	0.10	1.41	0.160	1	
log10_GDF15	3 vs 0	-0.04	0.07	-0.56	0.574	1	
log10_GDF15	4 vs 0	0.17	0.09	1.94	0.054	0.544	
log10_GDF15	2 vs 1	-0.02	0.10	-0.20	0.838	1	
log10_GDF15	3 vs 1	-0.19	0.08	-2.49	0.014	0.140	
log10_GDF15	4 vs 1	0.02	0.10	0.18	0.859	1	
log10_GDF15	3 vs 2	-0.17	0.09	-1.86	0.065	0.654	
log10_GDF15	4 vs 2	0.04	0.11	0.35	0.725	1	
log10_GDF15	4 vs 3	0.21	0.08	2.51	0.013	0.133	
log10_FGF_23	1 vs 0	0.27	0.10	2.62	0.010	0.100	
log10_FGF_23	2 vs 0	0.20	0.12	1.71	0.090	0.896	
log10_FGF_23	3 vs 0	-0.11	0.08	-1.26	0.209	1	
log10_FGF_23	4 vs 0	-0.07	0.11	-0.61	0.543	1	
log10_FGF_23	2 vs 1	-0.07	0.13	-0.52	0.604	1	

Biomarker	Comparison	Difference	SE	t	p_raw	p_bonf	Sig
log10_FGF_23	3 vs 1	-0.37	0.10	-3.88	0.000	0.002	**
log10_FGF_23	4 vs 1	-0.34	0.12	-2.81	0.006	0.058	
log10_FGF_23	3 vs 2	-0.31	0.12	-2.67	0.009	0.085	
log10_FGF_23	4 vs 2	-0.27	0.13	-2.00	0.048	0.478	
log10_FGF_23	4 vs 3	0.04	0.10	0.37	0.710	1	
log10_Syndecan4	1 vs 0	-0.12	0.10	-1.27	0.205	1	
log10_Syndecan4	2 vs 0	0.07	0.11	0.67	0.504	1	
log10_Syndecan4	3 vs 0	-0.25	0.08	-3.14	0.002	0.021	*
log10_Syndecan4	4 vs 0	-0.08	0.10	-0.73	0.468	1	
log10_Syndecan4	2 vs 1	0.20	0.12	1.65	0.102	1	
log10_Syndecan4	3 vs 1	-0.13	0.09	-1.38	0.169	1	
log10_Syndecan4	4 vs 1	0.05	0.11	0.42	0.675	1	
log10_Syndecan4	3 vs 2	-0.32	0.11	-2.98	0.003	0.035	*
log10_Syndecan4	4 vs 2	-0.15	0.13	-1.18	0.239	1	
log10_Syndecan4	4 vs 3	0.17	0.10	1.76	0.080	0.804	

B. Feature Importance Analysis:

To better understand what defines the patient clusters identified by the GCNN model, we trained a Random Forest classifier to predict which cluster each patient belongs to using the original features. Figure S4 shows the top 15 features, ranked according to how important they are to distinguish between groups. The results show that mean pre-treatment T1 (meanpreT1) is the most important feature for cluster assignment. This is followed by several measures related to the size and function of the left ventricle, such as LV end-systolic volume (LVESV) and LV end-diastolic volume (LVEDV), including their indexed forms (based on body surface area and height). Measures related to change to quality of life, such as MLWHFPhysScore and TotalMLWHFScore, were also found to be important. Other contributing features include post-treatment T1 values, global longitudinal strain (GLS absolute), blood T1 measurements, and some blood biomarkers like TIMP4, Syndecan-4.

Overall, the results suggest that the clusters are mainly distinguished by a combination of tissue characteristics (from T1 mapping), heart structure and function, and biomarkers linked to tissue remodeling. This indicates that the clusters reflect broader differences in heart structure and tissue condition, rather than being driven by a single clinical factor. This Post hoc feature importance analysis further indicated that several of these variables, particularly myocardial T1 measures, ventricular volume indices, and fibrosis-related biomarkers, contributed substantially to cluster discrimination, providing empirical support for their inclusion in graph construction.

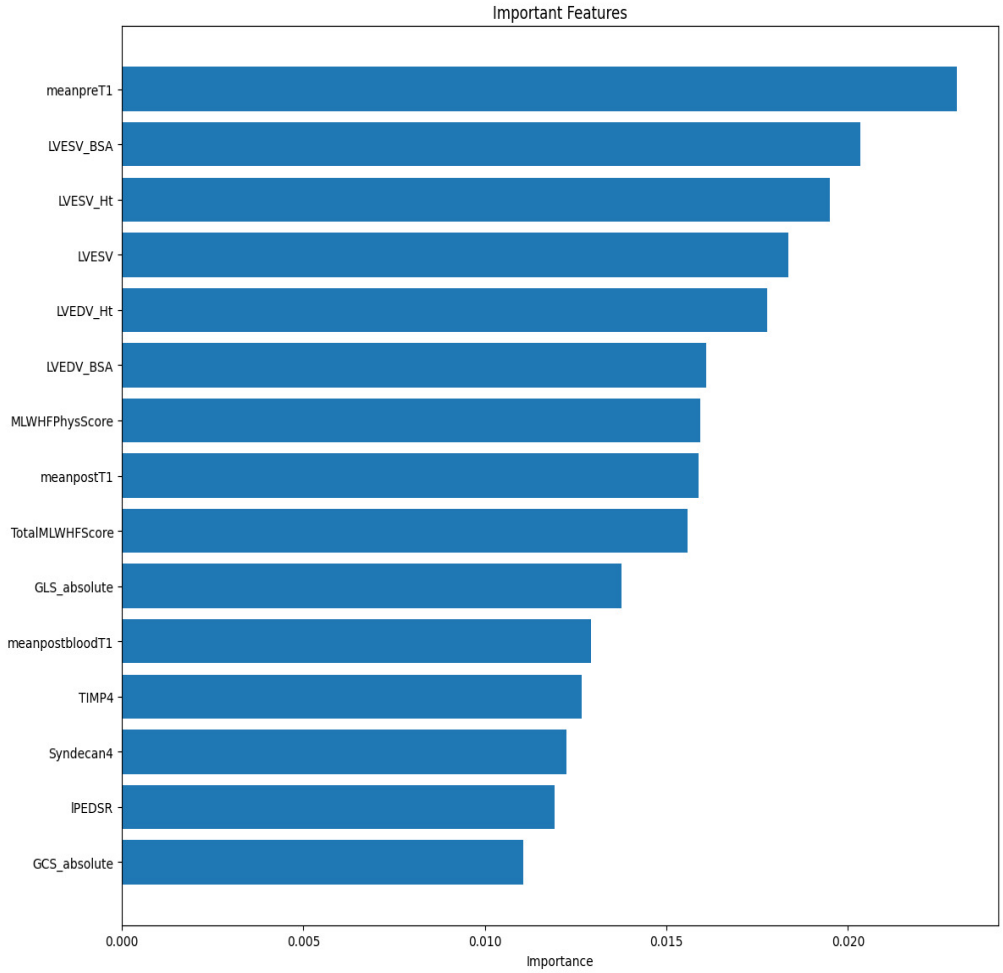


Fig. S4. Feature importance ranking obtained from a Random Forest classifier.

5. TRANSDUCTIVE VS. INDUCTIVE LEARNING PROTOCOL

Table S3. Comparison of the transductive and inductive learning protocols.

Experimental stage	Transductive Learning (Table IV)	Inductive Learning (Table VII)
Patients represented as graph nodes	All patients	All patients
Initial graph construction	Graph constructed using all patients based on cosine similarity.	Graph initially constructed using all patients based on cosine similarity.
Edges available during GCNN training	All graph edges are retained.	Only edges connecting training patients are retained; all train–test edges are removed.
Message passing during training	Across the complete patient graph.	Restricted to the training subgraph only.
Outcome labels used for training	Only labelled training nodes contribute to the supervised loss.	Only training-node labels contribute to the supervised loss.
Availability of test patient features during training	Yes, test nodes are present as unlabelled nodes.	Test node features exist in the feature matrix but cannot influence training because they are disconnected from the training graph.
Can information propagate from test patients to training patients?	Yes, through graph connectivity during message passing.	No, because train–test edges are removed.
Testing protocol	Predictions are generated on nodes within the same graph used during training.	Test embeddings are generated using a separate test graph containing only test–test connections.
Primary purpose	Compare graph architectures under full-graph availability.	Estimate performance on previously unseen patients under an inductive setting.

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